

The National Football League (NFL) is America's most popular and viewed league, and with only 17 regular-season games played every season per team, each win is crucial. The salary cap in this league is incredibly intricate and complex, with teams having a limited amount of funds to allocate to players' contracts. Being financially smart is incredibly important to success.

It is a well-known fact that the quarterback is the most important position on a football team, but there are five men, referred to as the Offensive-Line or O-Line (OL), in front blocking for him every play, so he is able to perform. This poses the question: Is investing in offensive linemen beneficial to winning football games? Given that there is a strict salary cap and teams must allocate funds responsibly under a fixed budget constraint, it is incredibly important that teams invest this limited amount of money properly. Every team is trying to win, and with only 17 games to do so, there is little room for error. I hypothesize that since the offensive line plays an irreplaceable role in protecting the quarterback, NFL teams that spend more on their offensive lines experience more wins in a given season. This is because with defenses trying to hit and attack the quarterback as quickly as possible, every tenth of a second could be the difference between a touchdown and an interception. Even if your quarterback is exceptional, if they have no time to play before they are tackled, then it severely hinders their ability to play well. Contrarily, average quarterbacks with great offensive lines will be able to have more time to throw the football each play, resulting in a better overall production.

In an unrestricted RCT scenario, I would reassign every offensive lineman to a random team, with all 32 teams forming my population. Some teams in this scenario would have very expensive and skilled offensive lines, while other teams would not be spending much at all, but would encounter worse play. The teams randomly assigned with overall more expensive O-Lines would be my treatment group, while the teams with relatively cheap O-Lines would be the

control group. All other aspects of players and staff would remain unchanged to single out the results that offensive lines have on teams winning games. This experiment would allow for a clear outcome of interest, in terms of the number of wins each NFL team achieves. Since only the offensive lines for each team changed, the difference in average wins between the treatment and control groups would be a causal effect of O-Line investments.

Although the RCT above is impossible, it is possible to compare teams' spending on an Offensive Line while also observing their win totals, using a team in an NFL season as my unit of observation. A regression to represent this is $Wins_{it} = \alpha + \beta OLspend_{it} + \epsilon_{it}$, with i being a given team and t being the given season. $Wins_{it}$ represents the team's total wins, and $OLspend_{it}$ represents the amount spent in millions on the offensive line in a given season. β can be interpreted as the effect on every million dollars spent on the OL. This regression is similar to my RCT experiment because offensive line spending is represented as a variable and is the only factor being directly analyzed.

My current regression suffers from Omitted Variable Bias (OVB), so β isn't a full causal effect. One evident example of OVB is the relationship between the most important player on the team, the quarterback, and the offensive line. A team that has a great quarterback will want to make sure he is well protected, which inclines them to spend more on an offensive line. In the regression above, offensive line spending inaccurately is credited for wins created by the quarterback. This results in β being upward-biased in this example, as an offensive line will be made to appear better than they actually are. Since data on [positional spending](#) is available, quarterback spending can also be added to the regression to reduce bias.

A second example of OVB could be the overall quality of a roster. Not all teams are constructed to win in the present season, and overall spending fluctuates by millions of dollars.

Because of this, some teams will spend more overall than others, including on the OL, and thus be better teams. This again results in β being upward biased, as wins attributed to teams being overall better than others could incorrectly depict offensive line spending as the sole reason. Data on teams' [overall spending](#) is reported, allowing for this to also be added to the regression.

My regression relies on the allocated spending to the offensive line position group, which can easily be found in this [dataset](#). This showcases each team's spending per position group and total on offense and defense. This allows for my main regression to have data, as well as my first case of OVB. Furthermore, a different [dataset](#), focused on overall team spending, allows for my second case of OVB to be accounted for. The dependent variable wins is easily accessible by searching for the NFL standings for any given season. Combining these two datasets provides the information I need for my regression to construct a complete panel dataset, while reducing the OVB present in my initial regression.

I would add team fixed effects in my regression, due to other factors like coaching or ownership quality and team culture, as in some teams being historically bad or good. These effects don't change from year to year but affect offensive line spending (frugal teams) and win totals, resulting in a correlation between OL spending and ϵ_{it} . By creating a team fixed effect, I would be able to compare each team to itself rather than other teams that are run completely differently. Implementing this would further allow me to reduce OVB and make the end regression more accurate.

Data sets:

Positioning Group Spending: <https://overthecap.com/cash-spending>; Total Team Spending: <https://overthecap.com/cash-spending>; Team Win Totals: <https://www.nfl.com/standings/>